

Abstract

Project Title

AI Learning Assistant

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The **AI Learning Assistant** is an intelligent educational application designed to provide personalized learning support by understanding the emotional state of students. Traditional learning platforms offer the same guidance to every learner without considering their emotional condition, which can affect motivation and learning efficiency.

This project combines **Artificial Intelligence**, **Natural Language Processing**, and **Google Gemini AI** to analyze user input, identify emotions such as **Happy, Sad, Angry, Stressed, Anxious, Motivated, Confused**, and **Neutral**, and generate personalized study guidance accordingly. The application is developed using **Python** and **Streamlit**, providing an interactive and user-friendly interface.

The system also records user interactions, detected emotions, confidence scores, and AI-generated responses in a CSV file. A dashboard is included to visualize learning trends and emotion distribution, helping users better understand their study patterns.

The AI Learning Assistant aims to create a more engaging, supportive, and adaptive learning environment by offering emotionally aware guidance. The modular architecture of the project also allows future enhancements such as voice emotion recognition, multilingual support, and cloud-based data storage.

Keywords

Artificial Intelligence, Streamlit, Google Gemini AI, Emotion Detection, Personalized Learning, Python, Educational Technology.